INVESTMENTS

Ninth Canadian Edition · Bodie, Kane, Marcus

CHAPTER 6

Capital Allocation to Risky Assets

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

Study Guide · Prepared for Personal Use

LO1 · Risk and Risk Aversion

Portfolio construction is usefully split into two tasks: (1) the **capital allocation decision** — how to divide the overall portfolio between a safe asset (e.g., T-bills) and a risky portfolio — and (2) **asset allocation / security selection** — how to construct the composition of the risky portfolio itself (the subject of Chapter 7 onward). This chapter tackles the first task.

Utility and Risk Aversion

To make the capital allocation decision rigorously, we need a way to rank portfolios that trades off expected return against risk. A widely used **utility function** (also used by the CFA Institute) assigns each portfolio a utility score based on its expected return and variance (see Formulas page). **A** is the investor’s coefficient of risk aversion: more risk-averse investors (larger A) penalize a given amount of variance more heavily.

A risk-free portfolio (variance = 0) receives a utility score exactly equal to its known return, since there is no risk penalty to subtract. As risk aversion A rises, an investor requires progressively more expected return to accept the same level of risk — which is precisely why more risk-averse investors optimally hold a smaller share of their wealth in risky assets (formalized later in this chapter).

> *Risk aversion is not directly observable, so financial advisers commonly use risk-tolerance questionnaires (asking about reactions to hypothetical losses, investment horizon, and income stability) to estimate a client’s effective coefficient A.*

LO2 · Capital Allocation Across Risky and Risk-Free Portfolios

Consider an investor allocating between a risky portfolio P and a risk-free asset. Let **y** be the fraction of investable wealth placed in P, and $(1 - y)$ the fraction in the risk-free asset. The resulting combination is the investor's **complete portfolio.**

At $y = 1$, the investor holds only the risky portfolio; at $y = 0$, only the risk-free asset. Values of y between 0 and 1 represent a **lending** position (holding some risk-free asset alongside the risky portfolio). Values of y greater than 1 represent **borrowing** at the risk-free rate to invest more than 100% of the investor's own capital in the risky portfolio — a levered position.

Expected Return and Risk of the Complete Portfolio

The complete portfolio's expected return is a simple weighted average of the risk-free rate and the risky portfolio's expected return. Its standard deviation is simply y times the risky portfolio's own standard deviation, since the risk-free asset contributes no variance and no covariance (see Formulas page for both relationships).

LO3 · The Risk-Free Asset

A truly risk-free asset must have a certain nominal return over the investor's holding period — in practice, a T-bill maturing at the end of the horizon is the standard proxy, since its price is fixed and there is essentially no default risk. (Note: even a T-bill is not risk-free in real terms, since unexpected inflation still creates real-return uncertainty — a nuance developed further in later chapters on bond and inflation risk.)

LO4 · Portfolios of One Risky Asset and a Risk-Free Asset

Plotting the complete portfolio's expected return against its standard deviation as y varies traces out a straight line connecting the risk-free point F (0% SD) and the risky portfolio point P — this line is the **Capital Allocation Line (CAL).** Its slope is: $[E(r_P) - r_f] / \sigma_P$ — exactly the **Sharpe ratio** of the risky portfolio (introduced in Chapter 5). Every point on the CAL represents the risk–return trade-off available at some allocation y, and the CAL's slope tells the investor exactly how much extra expected return is available per extra unit of risk.

> *Key insight: any investor, regardless of risk aversion, faces the same CAL (same risky portfolio P, same slope). What differs across investors is only where on that line they choose to sit — the choice of y. This separation of "which risky portfolio" from "how much risk to take" foreshadows the separation property developed fully in Chapter 7.*

LO5 · Risk Tolerance and Asset Allocation

Combining the utility function (LO1) with the CAL (LO4) lets us solve explicitly for each investor's **optimal complete portfolio.** The investor chooses y to maximize expected utility, $U = E(r_C) - \frac{1}{2}A\sigma^2_C$. Taking the derivative with respect to y and solving yields the **optimal risky-asset weight, y*** (see Formulas page) — directly proportional to the risky portfolio's risk premium, and inversely proportional to both the investor's risk aversion A and the risky portfolio's variance.

This result formalizes an intuitive pattern: more risk-averse investors (higher A) and/or riskier available portfolios (higher σ^2) both push y* lower, meaning a smaller allocation to the risky asset and a larger allocation to the safe asset.

Graphical Solution — Indifference Curves

The same optimum can be found graphically: an investor's **indifference curves** plot combinations of expected return and standard deviation that deliver equal utility (steeper curves for more risk-averse investors, since they require more extra return to compensate for a given increase in risk). The optimal complete portfolio sits at the point where the highest attainable indifference curve is exactly tangent to the CAL.

LO6 · Passive Strategies: The Capital Market Line

The risky portfolio P used to build the CAL can be chosen via **active** management (security analysis aiming to find mispriced assets) or a **passive** strategy that avoids direct or indirect security analysis entirely. A passive strategy might seem naive at first glance, but forces of supply and demand in large, competitive capital markets can make it a reasonable choice for many investors — a theme that will be developed fully once the Capital Asset Pricing Model (CAPM) is introduced in Chapter 9.

When the risky portfolio choice is a broad market-index fund, the resulting CAL is specifically called the **Capital Market Line (CML).** Chapter 5's historical data (an average risk premium of roughly 8.30% for large stocks, with a standard deviation around 20.28%) can be used to characterize the historical slope — and hence historical Sharpe ratio — of the CML using a broad equity index as the risky portfolio.

Fat Tails and the Passive Strategy

Because historical evidence (Chapter 5) suggests real-world returns can show fatter tails than the normal distribution predicts, investors facing this risk might rationally reduce their allocation to the risky portfolio (lower y) relative to what the simple mean-variance framework alone would suggest — extreme, rare, high-impact events ("black swans") are a real consideration even for a passive, diversified strategy.

Chapter 6 — Big-Picture Takeaways

- Portfolio construction splits cleanly into capital allocation (risky vs. risk-free) and the composition of the risky portfolio itself — this chapter covers the former.
- A quadratic utility function, $U = E(r) - \frac{1}{2}A\sigma^2$, lets investors rank portfolios by trading off expected return against risk, calibrated by risk-aversion coefficient A.
- The Capital Allocation Line traces every possible risk-return combination available by varying y; its slope equals the risky portfolio's Sharpe ratio.
- The optimal y^* is directly proportional to the risk premium and inversely proportional to risk aversion and variance — more risk-averse investors optimally hold less of the risky asset.
- The graphical solution (tangency between the highest indifference curve and the CAL) gives the same answer as the calculus-based y^* formula.
- When the risky portfolio is a passive, broad market index, the CAL becomes the Capital Market Line — a reasonable choice for many investors given how competitive large capital markets are.

The optimal risky-asset weight (F4) is the single most heavily tested calculation from this chapter.

F1 | Utility Score of a Portfolio

$$U = E(r) - \tfrac{1}{2} \times A \times \sigma^2$$

A = investor's coefficient of risk aversion; returns expressed as decimals. A risk-free portfolio ($\sigma = 0$) scores exactly U = its known return.

F2 | Expected Return of the Complete Portfolio

$$E(r_C) = y \times E(r_P) + (1 - y) \times r_f$$
$$= r_f + y \times [E(r_P) - r_f]$$

y = fraction of the complete portfolio invested in risky portfolio P. The second form highlights that expected return grows linearly in y at a rate equal to P's risk premium.

F3 | Standard Deviation of the Complete Portfolio

$$\sigma_C = y \times \sigma_P$$

The risk-free asset contributes zero variance and zero covariance, so total portfolio risk scales linearly with y.

F4 | Optimal Allocation to the Risky Asset (y*)

$$y^* = [E(r_P) - r_f] / (A \times \sigma_P^2)$$

Directly proportional to the risk premium; inversely proportional to risk aversion (A) and variance (σ^2). Derived by maximizing $U = E(r_C) - \tfrac{1}{2}A\sigma_C^2$ with respect to y.

F5 | Capital Allocation Line (CAL) Equation

$$E(r_C) = r_f + [\ (E(r_P) - r_f) / \sigma_P\ ] \times \sigma_C$$

The straight line through F (r_f, 0) and P ($E(r_P)$, σ_P) in expected-return/standard-deviation space. The bracketed term is the CAL's slope — the Sharpe ratio of P.

EXAMPLE 6.1 | Evaluating Investments by Utility Score

Given: Three investors with risk aversion $A_1 = 2$, $A_2 = 3.5$, $A_3 = 5$, all evaluating the same set of portfolios. Risk-free rate = 5%.

Because the risk-free portfolio has zero variance, all three investors assign it the same utility score:

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U(risk-free) = 0.05 for every investor, regardless of A
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For risky portfolios, the more risk-averse investors (higher A) assign progressively lower utility scores to the same portfolio. The textbook shows that a high-risk portfolio, H, is chosen only by the least risk-averse investor ($A_1 = 2$); a low-risk portfolio, L, is passed over even by the most risk-averse investor ($A_3 = 5$), because a portfolio with even more risk-reducing characteristics happens to also offer a better risk-return trade-off in this example. All three risky portfolios beat the risk-free alternative for these particular investors.

EXAMPLE 6.4 | Solving for the Optimal Complete Portfolio

Given: $r_f = 7\%$, $E(r_P) = 15\%$, $\sigma_P = 22\%$. Investor risk aversion A = 4.

Step 1 — Solve for y*:

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y* = (0.15 − 0.07) / (4 × 0.22²) = 0.08 / 0.1936 = 0.41
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This investor optimally places 41% of the complete portfolio in the risky asset and 59% in the risk-free asset.

Step 2 — Expected return and SD of the complete portfolio:

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E(r_C) = 7 + 0.41 × (15 − 7) = 10.28%
σ_C = 0.41 × 22 = 9.02%
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Step 3 — Confirm the Sharpe ratio matches P’s Sharpe ratio:

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[E(r_C) − r_f] / σ_C = 3.28 / 9.02 = 0.36 = [15 − 7] / 22 = 8/22 = 0.36  ✓
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Every point on the CAL — including this investor’s chosen complete portfolio — shares the same Sharpe ratio, since they all lie on the same straight line through F and P.

PRACTICE PROBLEM | Bow Valley Portfolio Solutions

Scenario

You are an advisor at Bow Valley Portfolio Solutions in Calgary. Your firm's recommended risky portfolio, P, has an expected return of 13% and a standard deviation of 25%. The current risk-free rate (T-bill yield) is 4%.

Part A — The Capital Allocation Line

(i) Calculate the Sharpe ratio of portfolio P.

(ii) Write the equation of the CAL relating $E(r_C)$ to σ_C.

(iii) If a client is willing to accept a standard deviation of 15% on their complete portfolio, what expected return does the CAL predict, and what value of y does this imply?

Part B — Utility Scores

Two clients are considering portfolio P versus staying entirely in T-bills. Client 1 has risk aversion $A = 3$; Client 2 has risk aversion $A = 6$.

(i) Calculate the utility score each client would assign to holding 100% of their wealth in portfolio P ($y = 1$).

(ii) Calculate the utility score each client would assign to holding 100% T-bills.

(iii) Based on utility scores alone, would either client prefer T-bills over full investment in P? Explain.

Part C — Optimal Capital Allocation

(i) Calculate the optimal y^* for Client 1 ($A = 3$).

(ii) Calculate the optimal y^* for Client 2 ($A = 6$).

(iii) For each client, calculate the expected return and standard deviation of their resulting optimal complete portfolio.

(iv) Verify that both optimal complete portfolios have the same Sharpe ratio as portfolio P itself, and briefly explain why this must always be true.

Part D — Leverage and Passive Strategy

A third, very aggressive client has risk aversion $A = 1.2$.

(i) Calculate this client's optimal y^*. Is it greater than 1? What does that imply about this client's position?

(ii) Explain in your own words what it would mean, practically, for this client to implement $y > 1$ using their brokerage account.

(iii) If Bow Valley builds portfolio P as a broad, passively managed market-index fund rather than through active security analysis, what would the resulting CAL be called, and why might a passive approach be reasonable given how competitive capital markets are?

Hints & Guidance

Part A tests LO4 and Formula F5. Part B tests LO1 and Formula F1 (mirror Example 6.1). Part C tests LO5 and Formulas F2–F4 (mirror Example 6.4 exactly, just with new numbers). Part D extends LO5–LO6 — remember $y^* > 1$ implies borrowing at the risk-free rate to lever up the risky position.